

Quiz 8 — The Transformer, Assembled

Context is the keys and values, match a spectrum to its tokens, fine-tune or not, and one diffusion step

Instructions: Answer all five questions — every part is answerable from the ideas and diagrams on the slides. Intuition first; no calculators needed.

Name: _____

Date: _____

1. Context is a blend of values. In attention, a token's output is a weighted average of the other tokens' vectors — and that output *is* the “context” it gathers. The three roles: the **query** asks, the **keys** set the weights, the **values** carry the content.

(a) Which of **query**, **key**, or **value** carries the actual content that ends up in a token's output — its context? → _____

(b) A peak attends over four peaks with weights 0.1, 0.7, 0.15, 0.05. Its new context vector is built mostly from which peak's value? → _____

2. Cross-attention: the encoder's K,V is the context. In an encoder–decoder (e.g. Casanovo: spectrum in, peptide out) the **encoder** reads the whole spectrum once and turns it into a set of **keys and values** — its memory of the spectrum. The **decoder** then writes the peptide one residue at a time, each step sending a query into that encoder K,V (*cross-attention*).

(a) As the decoder writes residue 1, then 2, then 3, . . . , is the encoder's K,V *recomputed every step* or *computed once and held fixed*? What does it represent?

(b) Suppose that halfway through writing the peptide you **swap in the K,V from a different patient's spectrum**. What happens to the rest of the peptide, and why?

3. Match each data type to a tokenization strategy. A transformer eats a sequence of vectors, so MS data must be turned into tokens. Match each data type to the one strategy that fits it best (each used once), using the decision chart from the slides.

(a) An **imaging-MS image** (a 2D picture, one spectrum per pixel) → _____

(b) A **raw binned spectrum** (a dense 1D vector of ~6,000 intensities) → _____

(c) A **peak list** (a sparse set of m/z + intensity peaks) → _____

Word bank: chunk into patches + linear projection (ViT) · CNN front-end · peak-as-token (Casanovo)

4. Fine-tune, or train from scratch? You need a classifier for a new MS assay. You have **800 labeled spectra**, and access to a transformer someone **pretrained on 2 million spectra**.

(a) Would you *train a new model from scratch* on your 800 spectra, or *fine-tune* the pretrained one? → _____

(b) In one sentence, why?